Jacob D. Holt, Ph.D. Candidate

jacob.d.holt.gr@dartmouth.edu
https://jakeh15.github.io/

Education

2020 – Present **Dartmouth College**
Ph.D. in Biology

2017 – 2020 **Georgia Institute of Technology (GPA: 3.33/4.0)**
B.S. in Biology with Research Option
Thesis title: *Effect of Hydrodynamic Regime on Snowflake Yeast Evolution*

2016 – 2017 **Georgia Southern University (GPA: 4.0/4.0)**

Publications

In press

1. G. Doing, A. J. Lee, S. L. Neff, *et al.*, "Computationally efficient assembly of pseudomonas aeruginosa gene expression compendia," *mSystems*, vol. 8, no. 1, J. A. Gilbert, Ed., e00341–22, Feb. 23, 2023, ISSN: 2379-5077. DOI: 10.1128/msystems.00341-22. (visited on 03/13/2024).

Under revision at Current Bio

1. J. D. Holt, D. Schultz, and C. D. Nadell, *Dispersal of a dominant competitor can drive multispecies coexistence in biofilms*, Nov. 28, 2023. DOI: 10.1101/2023.11.28.569042. (visited on 03/13/2024).

In prep

1. J. D. Holt, J. Peng, A. B. Dalia, and C. D. Nadell, *Adherence and recombination of dna on chitin flakes*, Mar. 13, 2024. URL: http://biorxiv.org/lookup/doi/10.1101/2023.11.28.569042 (visited on 03/13/2024).

Presentations

Talks

2024 Biofilm ecology of *V. cholerae* on chitin. M2P2 Retreat Lake Morey, VT.

2022 Mediating social microbes. Dartmouth College 3MT.

Selected Poster Abstracts

2024 Competitive advantage of parental lysogens in biofilms.
M2P2 Retreat Lake Morey, Vermont
J. Peng, **J. Holt**, A. Dalia, and C. Nadell

2023 Effect of predation on chitin degradation by *V. cholerae*. MCB Retreat.
MCB Retreat Shillington, New Hampshire
J. Holt, D. Schultz, D. Kadouri, and C. Nadell

Dispersal of a dominant competitor can modulate coexistence in multispecies biofilms.
GRC Microbial Population Dynamics Proctor, New Hampshire
J. Holt and C. Nadell

2022 Dispersal of a dominant competitor can modulate coexistence in multispecies biofilms.
ASM Biofilms Charlotte, NC.
J. Holt and C. Nadell

Presentations (continued)

- Growth of *E. coli* and *E. faecalis* on a titanium bead model.
  ASM Biofilms Charlotte, NC.
  M. Bond, **J. Holt**, and C. Nadell
- Microscale biofilm architecture of *Candida albicans* biofilms
  M2P2 Retreat Lake Morey, VT.
  S. Kassetty, **J. Holt**, D. Andes, D. Hogan, and C. Nadell
- Biofilm detachment as a passive method to survive bacterial predation
  M2P2 Retreat Lake Morey, VT.
  A. Plessier, **J. Holt**, D. Kadouri and C. Nadell

2019
- Effect of hydrodynamic regime on snowflake yeast evolution.
  Evolution of Complex Life Conference Atlanta, GA.
  J. Holt, W. Ratcliff, and P. Yunker

Awards

2020
- **President's Undergraduate Research Award**
  Funding awarded by the Georgia Institute of Technology for undergraduate thesis project.

2016
- **Presidential Scholar**
  Awarded by Georgia Southern University.

2016-2020
- **HOPE Scholar**
  Tuition scholarship awarded by the University System of Georgia.

Professional Work Experience

2021
- **Genetics teaching assistant**
  Biological Sciences, Dartmouth College Molecular and Cellular Biology Ph.D. Program

2019
- **Undergraduate research assistant**
  Yunker Lab, Georgia Institute of Technology

2016 – 2017
- **Math and science tutor**
  Armstrong TRiO, Georgia Southern University

Committees and Service

Dartmouth College

2023 – 2024
- **Biology student representative to the graduate committee**
  Dartmouth College Molecular and Cellular Biology Ph.D. Program
  Responsibilities included; serving on the admissions committee, organizing recruitment weekends, organizing the MCB retreat, and voting on elective courses and faculty appointments to MCB.

2021 – 2023
- **Peer mentor**
  Dartmouth College Molecular and Cellular Biology Ph.D. Program

2020 – 2022
- **Peer research mentor**
  Dartmouth College Geisel School of Medicine.

Georgia Institute of Technology

2018-2020
- **Undergraduate research ambassador**
  Georgia Institute of Technology Vertically Integrated Projects.

Committees and Service (continued)

Non-Professional

2019-2020
- **Secretary**
  Georgia Institute of Technology Cycling Club.
- **Peer mentor**
  Mission Source Devo Junior Cycling Team.

2018-2020
- **Student representative**
  USA Cycling South Eastern Collegiate Cycling Conference.

2018-2019
- **Vice president**
  Georgia Institute of Technology Cycling Club.

2017-2018
- **New member coordinator**
  Georgia Institute of Technology Cycling Club.

Society Memberships

2021 – Present
- **American Society of Microbiologists**